Cellular and Medical Immunology: Tolerance and Autoimmunity Hypersensitivity Disorders

Integrated Biomedical and Clinical Sciences OMFS 513

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California Northstate University College of Dental Medicine

Mission: To advance the science and art of health through education, service, scholarship and social accountability

Learning Objectives

Learning Outcomes

1. Describe the mechanism of tolerance and autoimmunity.
2. Compare and contrast different types of hypersensitivity reactions.
3. Describe anaphylaxis.
4. Describe management strategies and anti-histamines

Outline

Tolerance and autoimmunity

What Is hypersensitivity?

Type I hypersensitivity

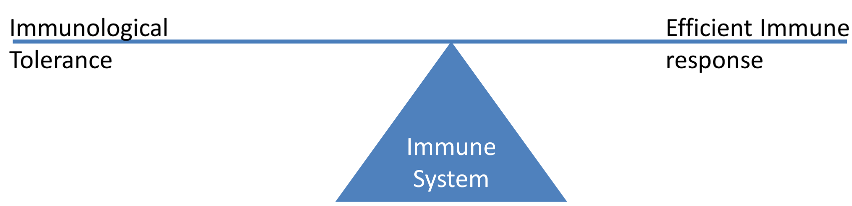
Type II hypersensitivity

Type III hypersensitivity

Type IV hypersensitivity

Management strategies

Tolerance and Autoimmunity


Immunological
Tolerance
Efficient Immune
response
Immune
System

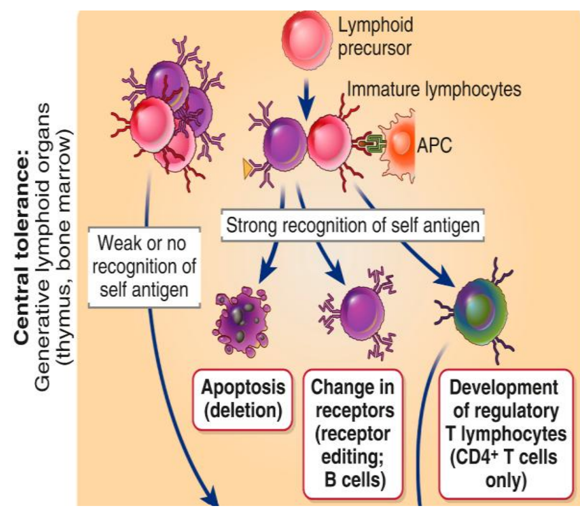
What is Immunological Tolerance?

Self-tolerance is important for the normal function of the immune system. To achieve self-tolerance, the immune system has two main strategies to eliminate self-reacting lymphocytes : **central and peripheral tolerance (location?)**.

Failure of self-tolerance can lead to -------------------------

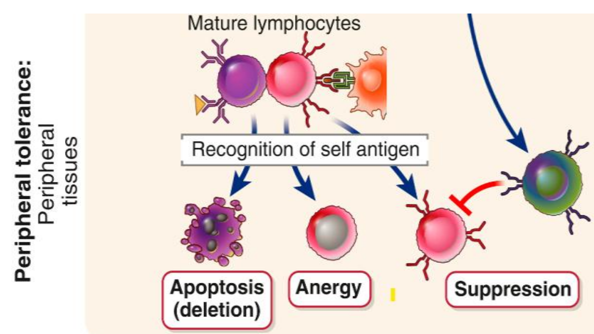
Central Tolerance

- Via negative selection, bone marrow (B-cell) and thymus (T-cells).
- Immature self-reactive lymphocytes (high avidity tolerogen recognition) are inactivated via:
  - Clonal deletion: lymphocyte death (B- & T-cell)
  - Receptor editing: replacement of self-reactive antigen receptor with non-self-reactive receptor (B cells).
  - Development into a Treg (--- cells).



Peripheral Tolerance

- Inactivation of mature self-reactive lymphocytes in the peripheral tissues and lymphoid organs through:
  - suppression with regulatory T cells,
  - clonal anergy
  - peripheral deletion (apoptosis).



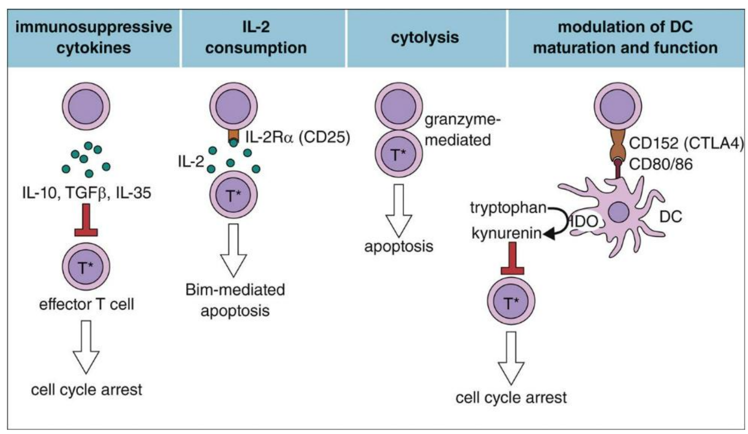
Suppression by regulatory T-cells (TREGs)

- Identifying markers: Tregs are $CD4^{+}$ $FoxP3^{+}$ $CD25^{high}$. FoxP3 and CD25 are essential for the generation & function of Tregs (patients with defects in *FoxP3* are at higher risk for autoimmune diseases).

- Main function: Act on different target cells (effector T cells and DCs, B cells, macrophages, NK cells…) to maintain self-tolerance and suppress immune response.

TREGs effector mechanisms:

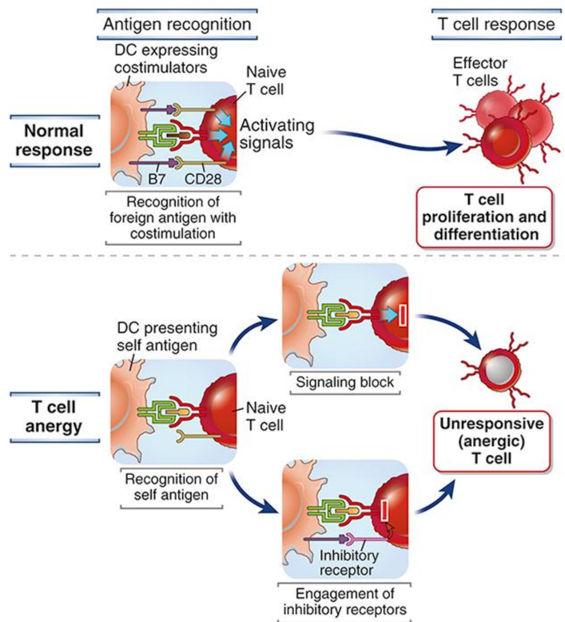
- Suppressive cytokines.
- Reduce the ability of APCs to stimulate the T-cells.
- IL-2 over-consumption
- Induce apoptosis.



Anergy

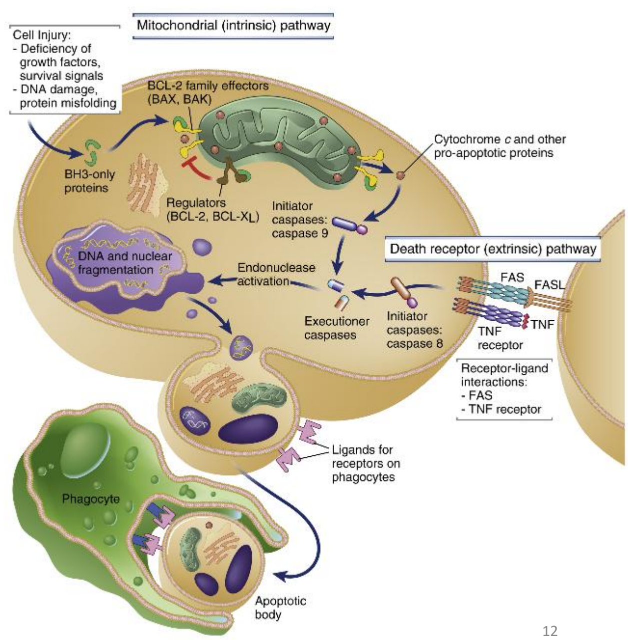
In anergy, a self-reacting B- or T-cell is deprived form co-stimulatory signals necessary for its activation. Instead, they engage co-inhibitory receptors that function to terminate T- cell responses. B-cells lose IgM and keep the un-responsive IgD.

Self-reactive cells do not die, but they become unresponsive to the antigen.



Peripheral deletion

- Self-reactive T-cell will be removed via induction of apoptosis (intrinsic or extrinsic pathway).
- Self-reactive B-cells won’t receive stimulatory signal from T-cell and removed by apoptosis (intrinsic pathway).



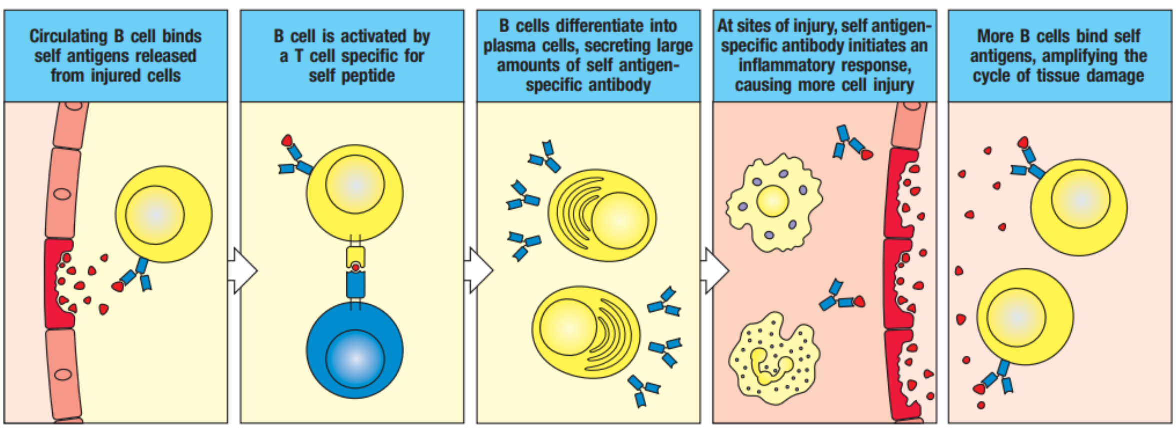
Failure: Autoimmune diseases

Autoimmunity is the breakdown in self-tolerance caused by inability to distinguish between self and non-self antigens or due to the misinterpretation of a self-component as dangerous, leading to an immune attack on host tissues.

Etiology: Genetic and environmental factors.
Category: Systemic (like RA, MS) or organ specific (like diabetes), depending on autoantigens distribution.
Course: most likely, Chronic and progressive.

- Self antigens that trigger these autoimmune reactions are persistent, and once an immune response starts, amplification mechanisms are activated that prolong and perpetuate the response.

Epitope Spreading



Provie positive feedback mechanism for autoimmune diseases

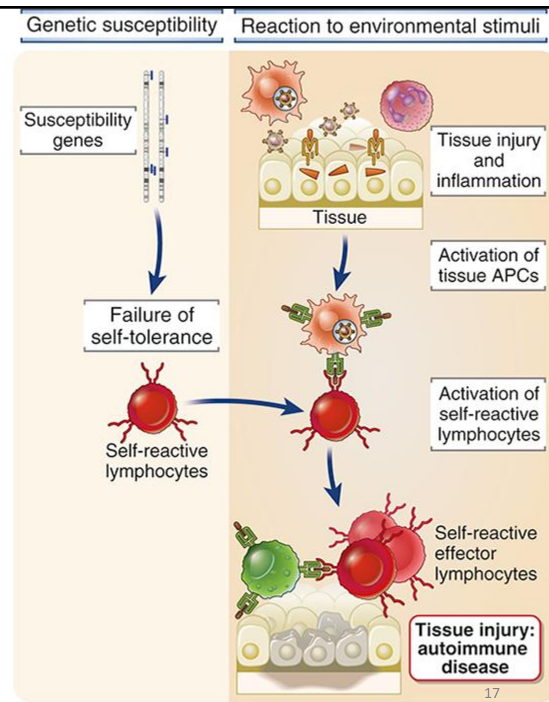
[Immunobiology (2005 edition) | Open Library]

How does self-tolerance fail in autoimmune disease?

Susceptibility genes may disrupt self-tolerance mechanisms.

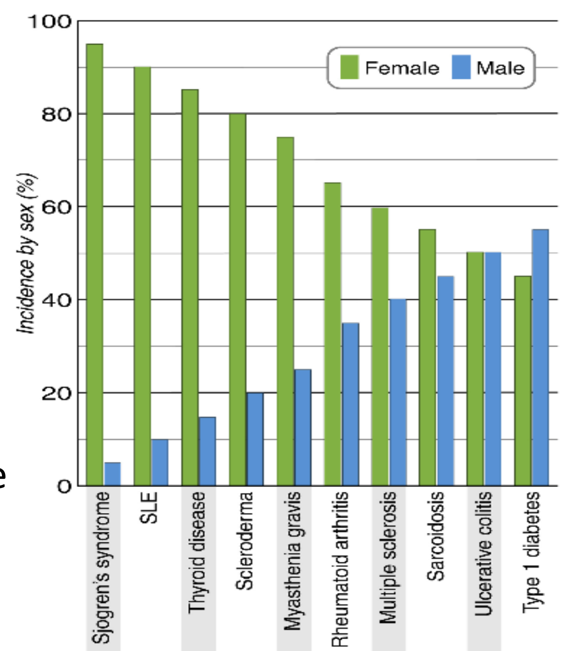
Susceptibility genes:

- *HLA* gene polymorphism. For example, certain *HLA* gene polymorphism has been reported with several autoimmune conditions like T1D, RA.
- Non-*HLA* genes. For example: Insulin in T1D (anti-insulin ab).



Anatomic alterations in tissues like ischemic injury, or trauma, may lead to the exposure of self-antigens that are normally concealed from the immune system.

Gender: Many autoimmune diseases have a higher incidence in women than in men. Hormones, sex-related immune difference….?



[The impact of sex on the immune system explored at the single-cell level: The American Journal of Human Genetics]

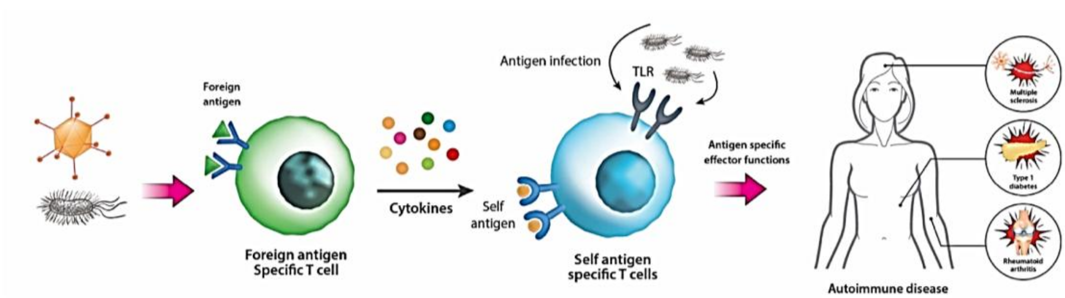
- **Environmental factors:**
  - Infections.
  - Drugs or chemicals can lead to failure of self-tolerance and activation of self-reactive lymphocytes.
  - Smoking and oxidative stress can increase the severity of autoimmune conditions

Role of infections in Autoimmunity

- Viral and bacterial infections may contribute to the development and exacerbation of autoimmunity. In most cases, autoimmunity does not target the infectious agent itself, but it is a consequence for damage induced by host immune responses (while fighting that infection)
- Infections may promote the development of autoimmunity by two principal mechanisms
  - ✓ bystander activation.
  - ✓ molecular mimicry.

Bystander activation:

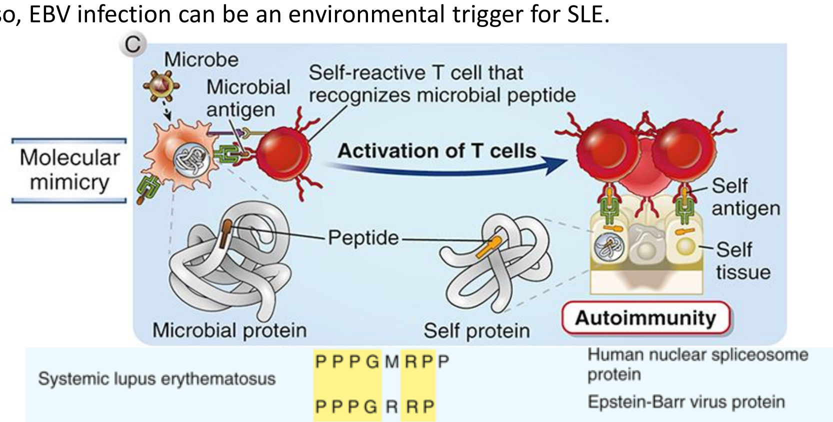
- Bystander T cell activation refers to the activation of T cells without antigen recognition.
- During infections, significant numbers of T cells can be activated secondary to the release of cytokines “cytokine-dependent manner” without activation of the T-cell receptors “T cell receptor-independent”.
- This can break the T-cell self tolerance & formation self-reactive T cells (not specific for the infectious pathogen). Instead, they will recognize and attack the self-antigens.



Emerging role of bystander T cell activation in autoimmune diseases (bmbreports.org)

Molecular mimicry:

Infectious microbes may contain antigens with high similarity to our self antigens. Example: *Strept pyogenes* can cause rheumatic fever. Why? Antibodies for S. pyogenes antigens (carbohydrate N-acetylglucosamine) can also recognize heart muscle proteins and drive complement activation. Also, EBV infection can be an environmental trigger for SLE.



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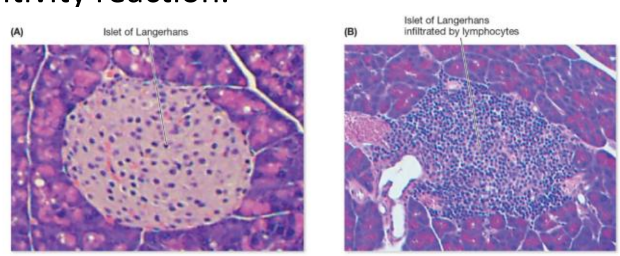
Autoimmune Conditions

- Thyroid:
  - Hashimoto thyroiditis: autoantibodies against thyroglobulin (precursor for T3/T4), attacking thyroid tissue & yielding hypothyroidism.
  - Graves Disease: thyroid-stimulating immunoglobulins (TSI) that overstimulate the thyroid, yielding hyperthyroidism. Classified as autoimmune and type II hypersensitivity reaction.
- Pancreas: type I diabetes
- Stomach: autoantibodies for intrinsic factor prevent vit. B12 absorption= pernicious anemia.



Type 1 Diabetes Mellitus

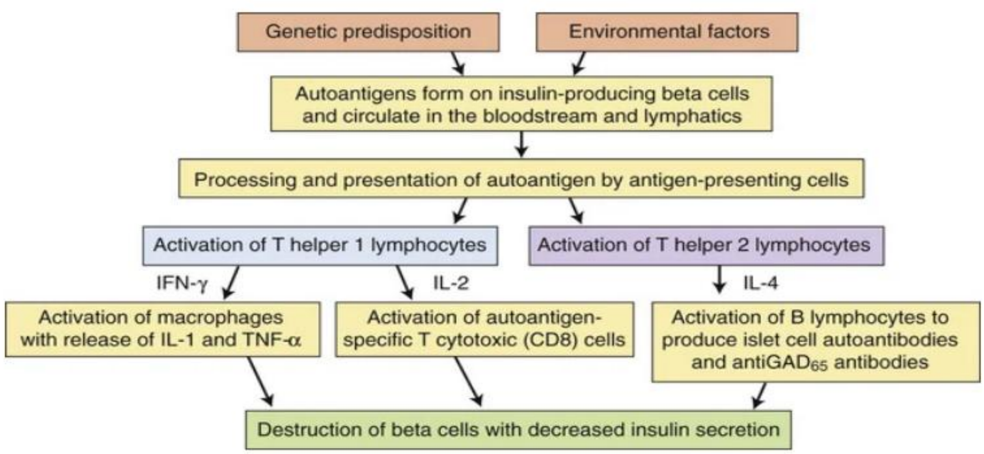


FIGURE 22.12 Pathophysiology of Type 1 Diabetes Mellitus. *GAD_{65}*, Glutamic acid decarboxylase; *IFN-γ*, interferon-gamma; *IL*, interleukin; *TNF-α*, tumor necrosis factor-alpha.

For illustration

Management

- Multiple strategies that are specific for disorder and stage of disease
  - Correct organ function.
  - Immunosuppression to control disease flare (short-course)
  - Biologics, like therapeutic antibodies to target specific elements of the immune system.

Hypersensitivity reactions

What is hypersensitivity?

- Healthy immune response is tightly regulated to prevent immune aggression and immune suppression pathways.
- Hypersensitivity = overreaction of immune system towards foreign benign antigens.
- Four different types of hypersensitivity reactions caused by either antibodies or T-cells. Hypersensitivities develop in two stages, a sensitization stage and an effector stage
- Hypersensitivity diseases tend to be chronic and progressive and pose major therapeutic challenges.

Causes of hypersensitivity diseases

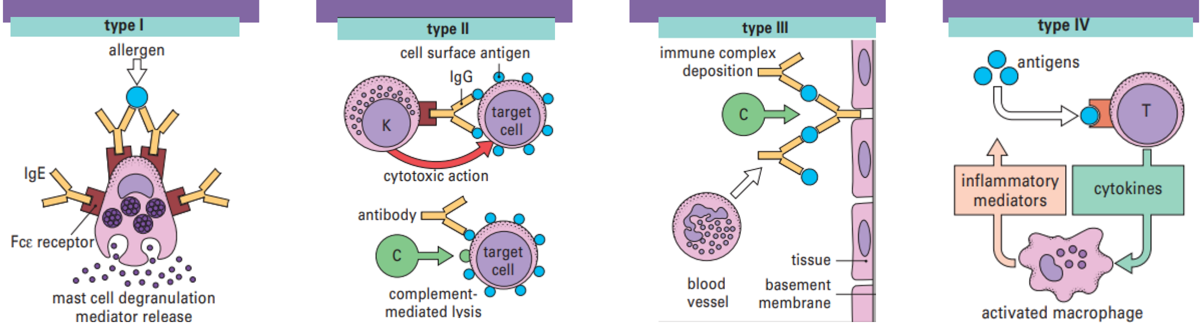
1. Reactions against self antigens. Failure of self-tolerance leads to autoimmunity.

2. Exaggerated reactions against microbes. Immune responses against microbial antigens may cause diseases if the reactions are excessive or in persistent infection.

3. Reactions against nonmicrobial environmental antigens like chemicals, pollens and food antigens (around 20 % prevalence).

Hypersensitivity reactions

Type I Hypersensitivity	Type II Hypersensitivity	Type III Hypersensitivity	Type IV Hypersensitivity



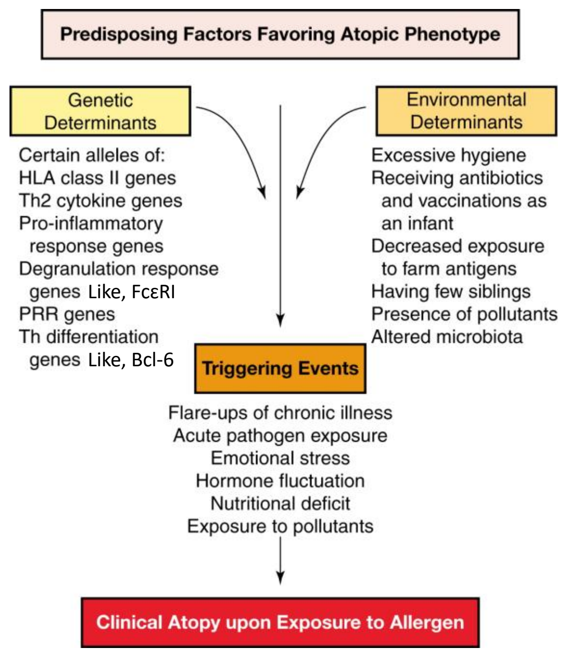
Type I hypersensitivity

Allergy is exaggerated immune response to an antigen regardless of mechanism. Named according to causative agent. For example, penicillin allergy, food allergy and bee sting allergy.

Atopy: genetic predilection of an individual to produce specific IgE antibodies and develop strong hypersensitivity response. Individuals with allergies to environmental stimuli like pollen and dust are termed atopic.

Anaphylaxis describes a severe, life-threatening allergic response.

What makes an individual allergic?



Nature of Allergens

- Allergens: antigens that lead to hypersensitivity and contribute to asthma rhinitis or food allergy.
- Most allergens are proteins of relatively small molecular weight. Also, chemicals bound to proteins (e.g: pollen, house dust mites, animal dander, foods, and drugs).
- Genetically predisposed individuals can make IgE in response to different allergen sources.
- Upon exposure to allergens, individuals with allergies can develop symptoms that include sneezing, wheezing and difficulty in breathing (asthma), conjunctivitis, or urticaria (hives).

What changes an antigen to an "allergen"?

1- Structural feature

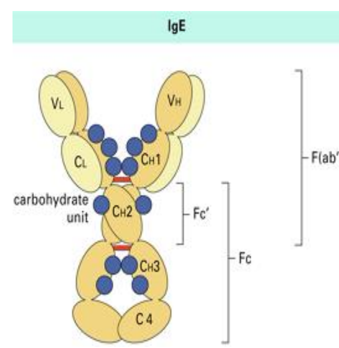
Structural resemblance with TLR ligands (for example Derp2, an allergen from the house dust mites resemble LPS and can bind to the lipid binding moiety of TLR4).

2- The ability to bind to lipids.

Binding lipids can help protect the antigens from digestion or from expulsion by the respiratory system.

Immunoglobulin E

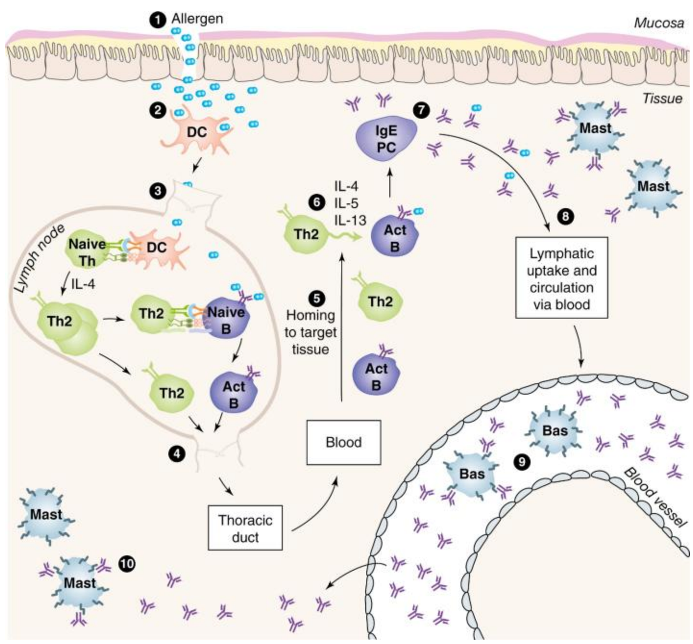
- IgE has a low serum concentration relative to the other antibodies. It contains the ε heavy chain.
- **IL-4** stimulate B cell antibody class switching to IgE.
- Imp. role in atopic conditions and contributes to the body's immune response to parasitic infections.
- Around 50 % of IgE circulates in blood. The remaining is bound to the mast cells and basophils receptors [(high-affinity IgE-specific receptor (FcεRI)]. Another low-affinity IgE receptor (FcεRII) is expressed on B cells and some myeloid cells.



Two stages:
Sensitization stage and Effector stage.

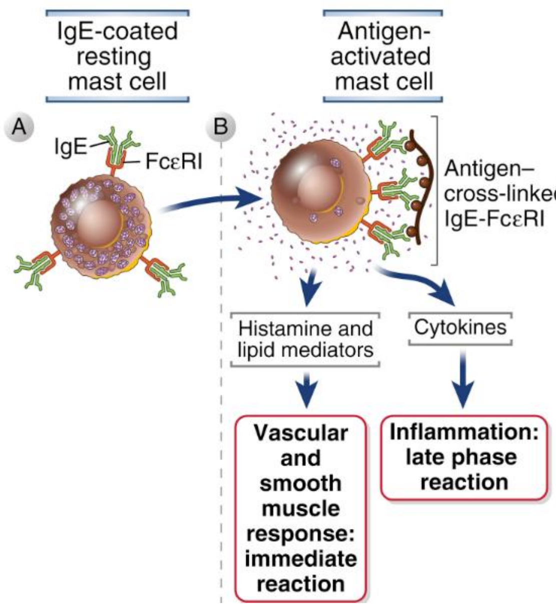
Sensitization stage

1- Allergen gain access.
2- DCs.
3- DCs-T cell interaction → Th2 (IL-4).
4- Th2 cells activate B cells.
5- Th2 and B cells arrive to injury site.
6- Th2 cells produce IL-4, 5 and 13. IL-4 induce class switching
7- IgE sensitize mast cells for future exposure.
8- Ig E enter the blood and sensitize basophils, eosinophils and mast cells away from the injury site by binding to their FCε receptor 1 (FCεR1).



The effector stage

Second exposure: allergen cross link the IgE-FCεRI on the sensitized mast cell leading to release of inflammatory mediators from the granules. The process of activation of granulocytes and release of its contents is referred to as “degranulation”



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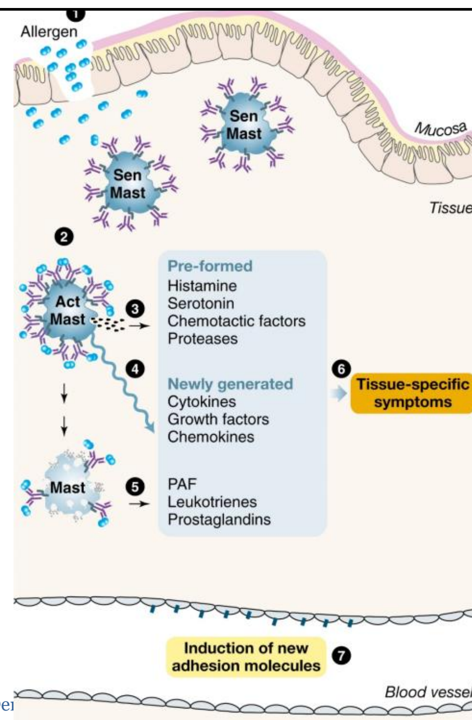
- Interleukin-13 is involved in mucus secretion and interleukin-5 recruits more leukocytes (eosinophils).
- In addition to IgE, mast cells, basophils, and eosinophils are essential components of allergic inflammation.
  - Mast cells and basophils are main source of histamine in the immune system. Activation of mast cells results in secretion of granule contents by exocytosis (degranulation), synthesis and secretion of lipid mediators (phospholipase A2), and synthesis and secretion of cytokines
  - Eosinophils secrete proinflammatory cytokines, peroxidase and major basic protein (50% of its secretion). The major basic protein has activity against parasites and can cause bronchial hyperresponsiveness in asthma.

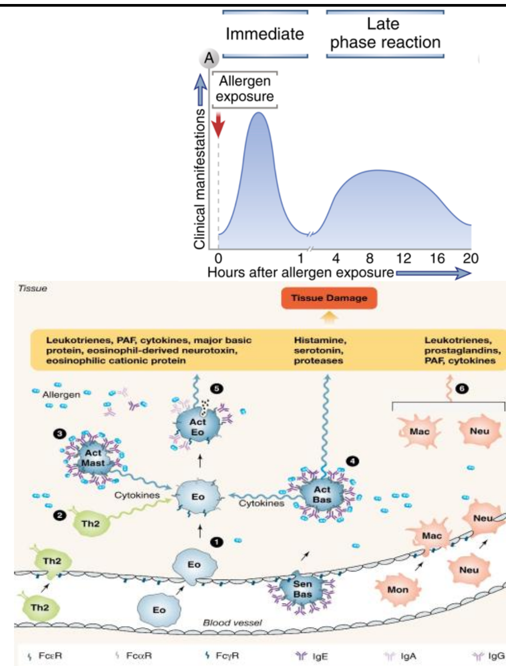
Phases of Effector Stage

1- Early response within minutes of allergen second exposure.
Results from the release of histamine, leukotrienes, and prostaglandins from local mast cells (collectively known as primary mediators) through degranulation. All these mediators produce the tissue specific symptoms that are typical of allergic reactions (hives, tingling of the lips, swelling).
The early response occurs within minutes from the second exposure. Therefore, anaphylaxis has a rapid onset after re-exposure.



2- Late-phase reaction.

- Occurs within few hours-24 h after the immediate phase of a type I hypersensitivity and gradually subsides. Mediated by inflammatory mediators released during first phase.
- Cytokines act on the endothelial cells and facilitate the influx of the second wave of inflammatory cells (including neutrophils, eosinophils, basophils, and helper T cells).
- Those cells further amplify the allergic response, produce leukotrienes and can cause tissue damage.



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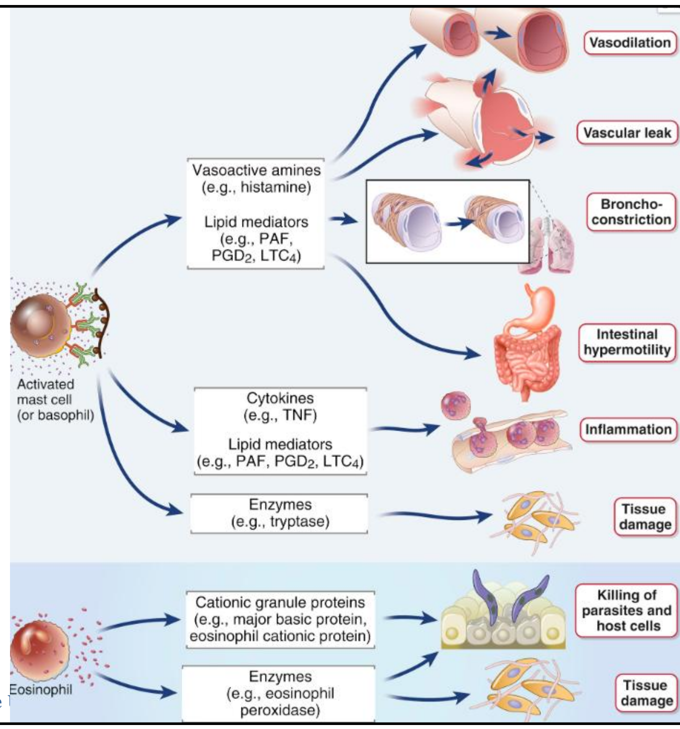
Inflammatory mediators and their biologic effects



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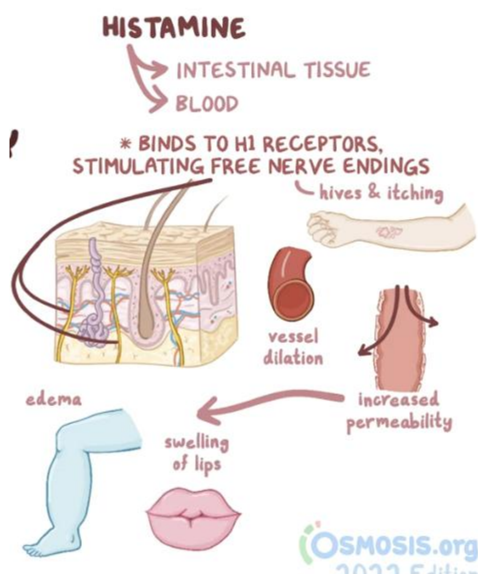
Food Allergy (IgE-mediated)

- Common food allergens: proteins in milk, eggs, peanuts, tree nuts, seafood, shellfish, soy, and wheat.
- Allergens are picked up by M-cell (antigen presenting cells in Peyer's patches and presented to Th cells (develop to Th2 and induce IgE).
- Release of histamine and other inflammatory mediators lead to vasodilation, swelling of lips, severe reactions lead to hypotension, bronchoconstriction and anaphylaxis.



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- Food allergies can also be non-IgE mediated:
  - E.g: food protein-induced enterocolitis syndrome (to milk) or celiac disease (to gluten).
  - The mechanism of the non-IgE mediated type is not well understood.
  - Symptoms occurs hours or days after exposure and they include GI symptoms like abdominal pain, constipation/diarrhea.

Anaphylaxis

- **International Consensus on Anaphylaxis (ICON)** described anaphylaxis as "a serious, generalized or systemic, allergic or hypersensitivity reaction that can be life-threatening or fatal".
  - Anaphylaxis is an acute, but severe allergic reaction.
  - Caused by the release of chemical mediators from mast cells and basophiles following degranulation. It can involve two or more organ systems.
  - Usually initiated by an allergen introduced directly into the bloodstream or absorbed from the gut or skin.
- A wide range of antigens has been shown to trigger this reaction in susceptible humans, including the venom from bee, wasp, hornet; drugs such as penicillin, insulin, and antitoxins; and foods such as seafood and nuts.

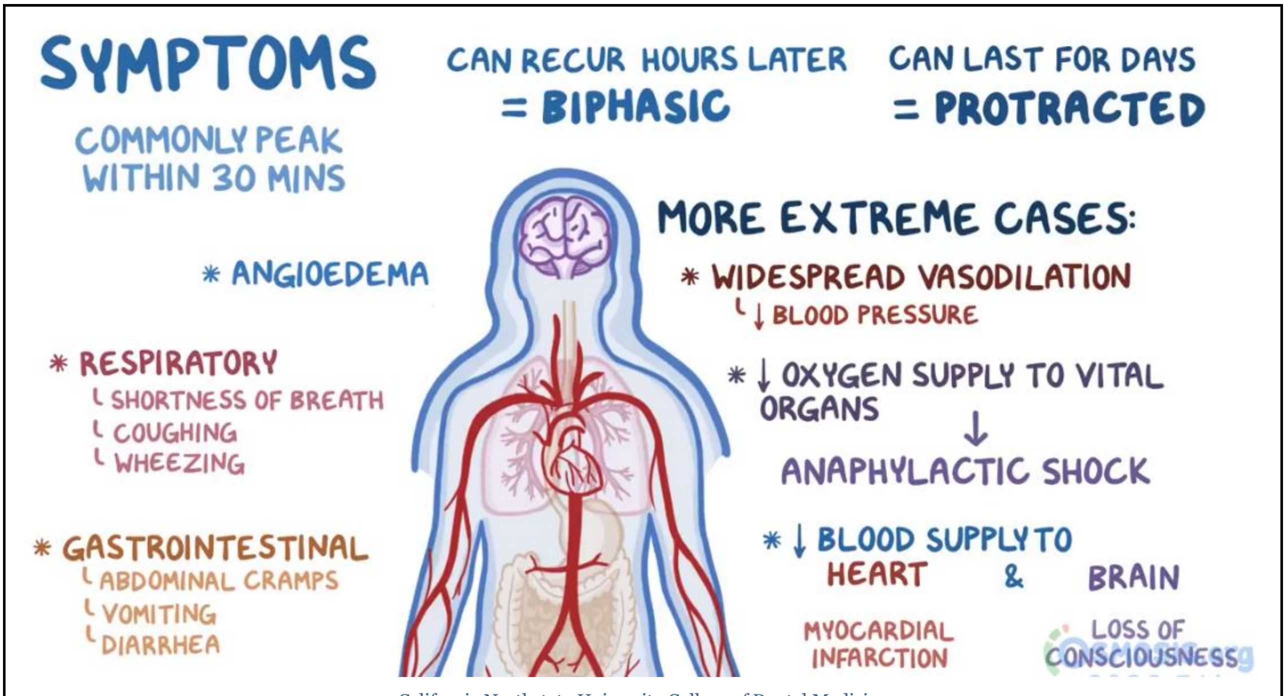

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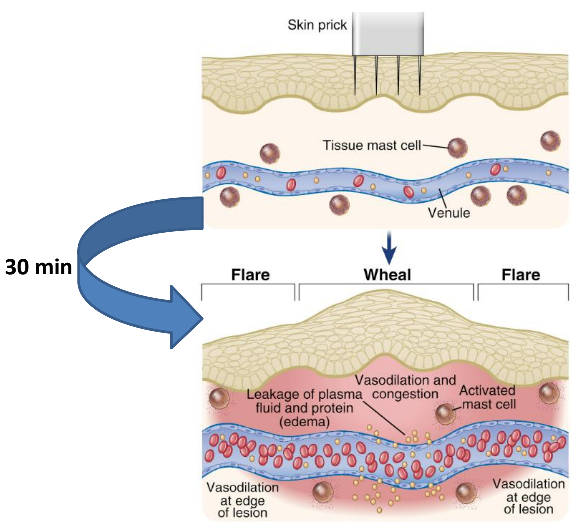
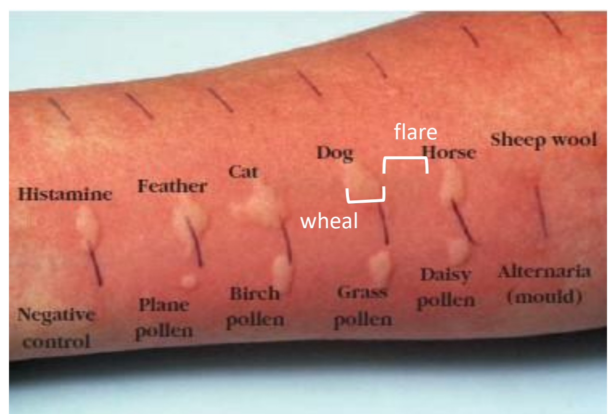
TABLE 15-1 Common allergens associated with type I hypersensitivity

Plant pollens	Foods
Rye grass	Nuts
Ragweed	Seafood
Timothy grass	Eggs
Birch trees	Peas, Beans
Drugs	Milk
Penicillin	
Sulfonamides	**Insect products**
Local anesthetics	Bee venom
Salicylates	Wasp venom
	Ant venom
Mold spores	Cockroach calyx
Animal hair and dander	Dust mites
Latex	
Foreign serum	
Vaccines	

Diagnostic Test for type-I hypersensitivity "Wheal and Flare skin response"





Management of Anaphylaxis

Epinephrine (EpiPen 0.3mg).

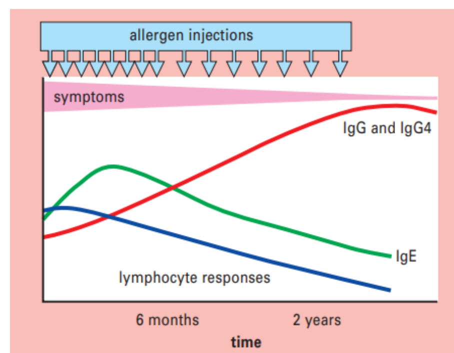
- Acts on α-adrenergic receptor to reverse the peripheral vasodilation and alleviate hypotension.
  - Epinephrine reduces vascular permeability and promotes vasoconstriction & peripheral vascular resistance.
- Acts on β adrenergic receptor to induce bronchodilation and increase the cardiac output.

Administer via intramuscular administration to the vastus lateralis of the quadriceps. May repeat after 5 min in alternate quadriceps.

Recent update on the management of anaphylaxis - PMC (nih.gov)

Desensitization: for Type I hypersensitivity

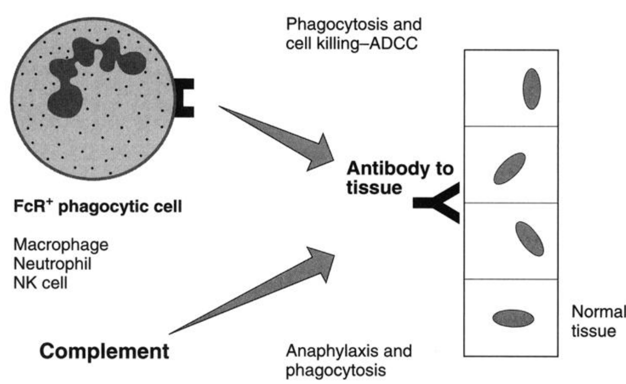
- Treating allergic patients with repeated increasing doses of allergens, is termed desensitization (aka, hyposensitization).
- Desensitization can reduce or even eliminate symptoms for months or years after the desensitization course is complete.



Desensitization can induce the production of IgG antibodies that can capture the allergen and prevent it from binding to individual's IgE antibodies.
Desensitization induce the formation of T_{reg} cells that inhibit the differentiation of TH2 cells and promote the Th1 differentiation.

Type II hypersensitivity

- Type II hypersensitivity is mediated IgG or IgM.
- Occur when IgG or IgM bind to a cell-surface antigen (exogenous or self-antigens) or extracellular matrix antigens leading to destruction of healthy cells or tissue damage.



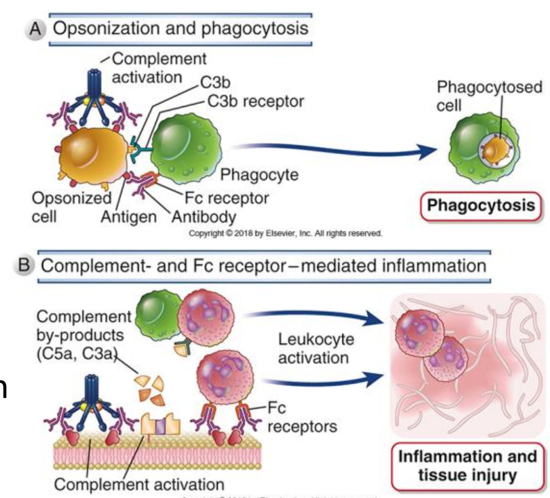
Effector Mechanisms of tissue/cell injury

A) Cytotoxic antibody disease

1-Opsonization.

2- Activation of classical complement pathway and complement mediated cytotoxic effect or inflammation and tissue injury.

3- Antibody-dependent cell-mediated cytotoxicity. NK cells recognize the Fc portion of the antigen-antibody complex and induce target cell apoptosis.



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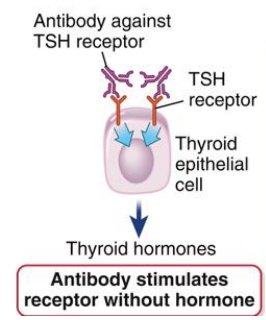
B) Non-cytotoxic leading to cellular dysfunction: loss or gain of function.

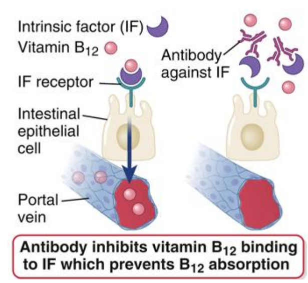
1- Grave's disease

Autoantibodies for TSH receptors stimulate the receptors and cause hyperthyroidism even in the absence of TSH.

2- Pernicious anemia

Antibodies against intrinsic factor interfere with vitamin B_{12} absorption resulting in its deficiency and causing pernicious anemia.





Antibodies that cause cell- or tissue-specific diseases are usually autoantibodies produced as part of an autoimmune reaction. However, antibodies are also produced against microbes or medications.

Type II hypersensitivity reactions in Cells

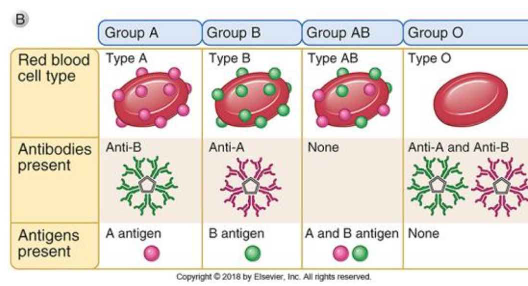
Hemolytic reaction:

1- Transfusion reactions

Blood groups must be matched for transfusion.

Antibodies directed toward ABO antigens are IgM antibodies.

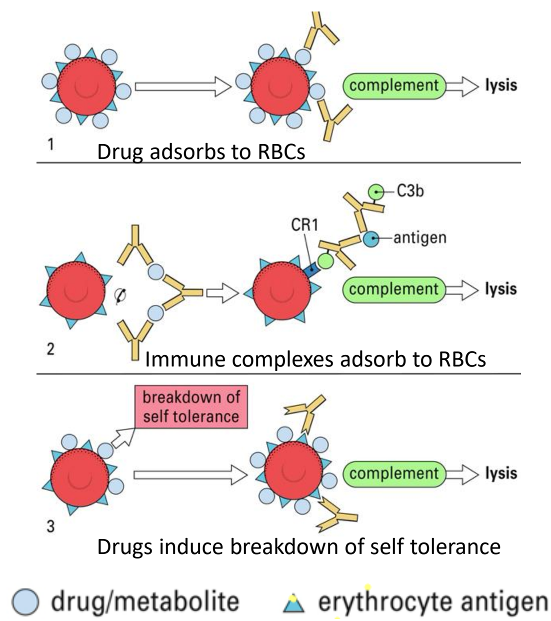
Clinical manifestations of transfusion allergic reactions include massive intravascular hemolysis of the transfused red blood cells by antibody plus complement.



2- Drug induced

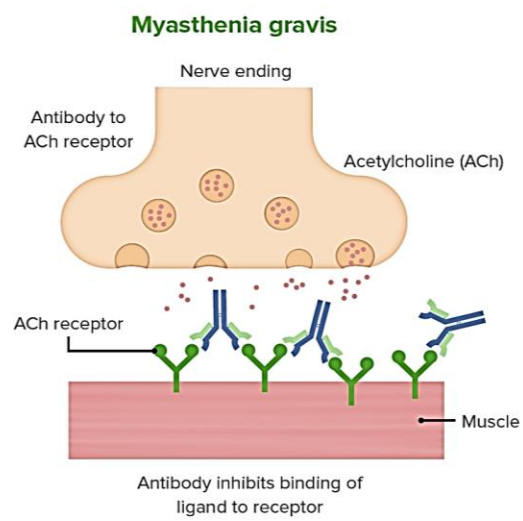
Certain drugs (like penicillin, quinine and sulfonamides) can cause hemolytic anemias by a type II hypersensitivity reaction (rare complication).

Drug binds RBCs and antibodies are produced against the drug. This activates the RBCs lysis via the activation of the complement system.



Type II hypersensitivity reactions in tissues

Myasthenia gravis is an autoimmune disease characterized by extreme muscular weakness. Autoantibodies to nicotinic acetylcholine receptors (neuromuscular junction) prevent the binding of Ach to its receptors leading to muscle weakness.



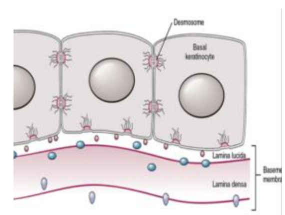
Pemphigus

Pemphigus vulgaris is a rare autoimmune condition that causes blistering of the skin and mucous membranes. Mean age of onset>50 Y.

Pathophysiology: Patients have autoantibodies (usually IgG or IgA) against desmoglein (adhesion proteins, components of desmosomes that form the intercellular junctions and mediate cell–cell adhesion).

Type II hypersensitivity disrupt cellular adhesion, leading to breakdown of the epidermis with separation of the superficial layers to form blisters and the release of proteases leading to tissue damage.

Treatment: Corticosteroids



URE 1-17 Cutaneous pemphigus vulgaris.

Pemphigus vulgaris of the lower lip.

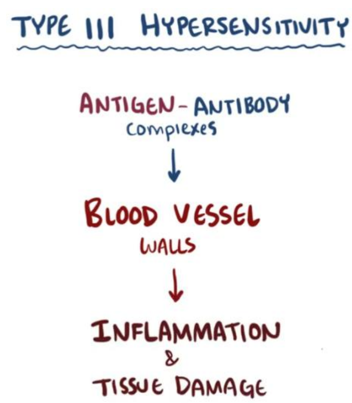
Examples of Antibody-Mediated Diseases (Type II Hypersensitivity)

Disease	Target Antigen	Mechanisms of Disease	Clinicopathologic Manifestations
Autoimmune hemolytic anemia	Red blood cell membrane proteins	Opsonization and phagocytosis of red blood cells	Hemolysis, anemia
Autoimmune thrombocytopenic purpura	Platelet membrane proteins (GpIIb : IIIa integrin)	Opsonization and phagocytosis of platelets	Bleeding
Pemphigus vulgaris	Proteins in intercellular junctions of epidermal cells (desmogleins)	Antibody-mediated activation of proteases, disruption of intercellular adhesions	Skin vesicles (bullae)
Vasculitis caused by ANCA	Neutrophil granule proteins, presumably released from activated neutrophils	Neutrophil degranulation and inflammation	Vasculitis
Goodpasture syndrome	Protein in basement membranes of kidney glomeruli and lung alveoli	Complement- and Fc receptor–mediated inflammation	Nephritis, lung hemorrhage
Acute rheumatic fever	Streptococcal cell wall antigen; antibody cross-reacts with myocardial antigen	Inflammation, macrophage activation	Myocarditis, arthritis
Myasthenia gravis	Acetylcholine receptor	Antibody inhibits acetylcholine binding, down-modulates receptors	Muscle weakness, paralysis
Graves disease (hyperthyroidism)	TSH receptor	Antibody-mediated stimulation of TSH receptors	Hyperthyroidism
Pernicious anemia	Intrinsic factor of gastric parietal cells	Neutralization of intrinsic factor, decreased absorption of vitamin B_{12}	Abnormal erythropoiesis, anemia

ANCA, Anti-neutrophil cytoplasmic antibodies; *TSH*, thyroid-stimulating hormone.

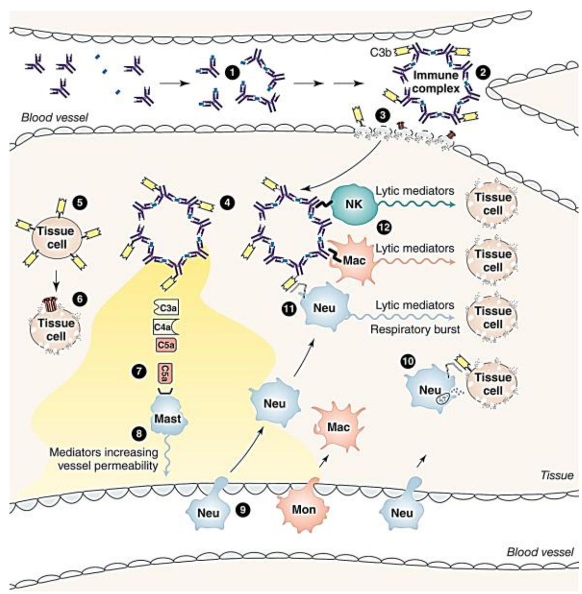
For illustration

Type III hypersensitivity



- Physiologically, antigen-antibody form an immune complex. These immune complexes are removed by the liver and spleen via several mechanisms.
- Type III hypersensitivity occur when soluble antigen binds IgG or IgM & form insoluble immune complexes.
  - Immune complexes persist and deposit in the blood vessel and cause systemic diseases, often preferentially involving the kidney (glomerulonephritis), joints (arthritis), and small blood vessels (vasculitis) because these are common sites for immune complex deposition.
- The deposition of these complexes initiates an immune response resulting in the recruitment of complement components and phagocytic cells.

- Tissue damage is further aggravated by activation of complement system, free radicals and proteases released by the activated neutrophils and macrophages at the injury site.
- Immune complexes can occur secondary to persistent infection or autoimmune disease.



- Site of deposition is determined by the antigen location. The site for the immune complex deposition determine the manifestation.
- Type II and type III hypersensitivity reactions have some similarities, and they are not mutually exclusive.

Three Categories of Immune Complex Disease

Cause	Antigen	Site of Complex Deposition
Persistent infection	Microbial antigen	Infected organ(s), kidney
Autoimmunity	Self antigen	Kidney, joint, arteries, skin
Inhaled antigen	Mould, plant or animal antigen	Lung

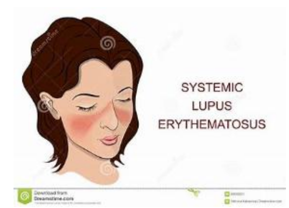
1- Poststreptococcal Glomerulonephritis:

Following streptococcal infection, Streptococcal cell wall antigens and antibody immunocomplexes bind to the basement membrane of the glomeruli and set up a type III response, which resolves as the bacterial load decreases.

2- Arthus reaction:

A sensitive individual may react to an allergen (insect bite, chemical, bacterial or fungal spore) with a rapid, localized type I reaction, which can be followed, some 4 to 10 hours later, by the development of an Arthus reaction, characterized by pronounced erythema and thrombosis which may lead to tissue necrosis.

3- Autoimmune diseases:



In systemic lupus erythematosus (SLE), there is persistent antibody responses to autoantigens. Autoantigen represent the nuclear antigens (histone, DNA or other nucleoproteins). Immune complexes (nuclear antigens and anti-nuclear antibodies) are deposited in the blood vessel of the kidneys, and skin of patients.

4- Serum Sickness

Can occur due to medications (penicillin and other antibiotics), vaccines, infections (Hep B), insect bites, use of monoclonal antibodies. Affected persons with generate antibodies towards the offending antigen. Deposition of immunocomplexes may lead to vasculitis, nephritis and arthritis.

TABLE 15-5	Examples of diseases resulting from type III hypersensitivity reactions.
Autoimmune diseases	
Systemic lupus erythematosus	
Rheumatoid arthritis	
Multiple sclerosis	
Drug reactions	
Allergies to penicillin and sulfonamides	
Infectious diseases	
Poststreptococcal glomerulonephritis	
Meningitis	
Hepatitis	
Mononucleosis	
Malaria	
Trypanosomiasis	

Type IV hypersensitivity
T-cell mediated

Type IV hypersensitivity or **Delayed-Type Hypersensitivity** (DTH), is T- cell mediated (vs type I-III antibody mediated).

Type IV reaction is initiated by T cells, and the delay is caused by time needed for the reaction to develop. Macrophages as the main innate cellular component in the site of inflammation.

- Initial sensitization by antigen: antigen-specific T cells are activated and clonally expanded (1-2 weeks)
- Effector stage: Memory T cells are stimulated to secrete a variety of cytokines/chemokines, including IFN γ which recruits and activates macrophages and other inflammatory cells. A DTH response is not apparent until an average of 24 hours after the second antigen encounter and peaks 48 to 72 hours after this stimulus.

Hallmarks of a type IV reaction:

1- Initiation by T cells.

2- Delay or lag period is required for the allergic reaction to develop. A period of 1-2 weeks in the sensitization state and 48-72 h in the effectors stage

3- Inflammation: cytokines and recruitment of macrophages as the main innate cellular component of the infiltrate that surrounds the inflammation site.

4- In some cases, cytotoxic T-cells can directly kill the cells.

Common examples include allergic contact dermatitis, granulomas (tuberculosis), and Crohn's disease. Also, certain medications (like penicillin) can cause type IV reactions.

APC activates T cell

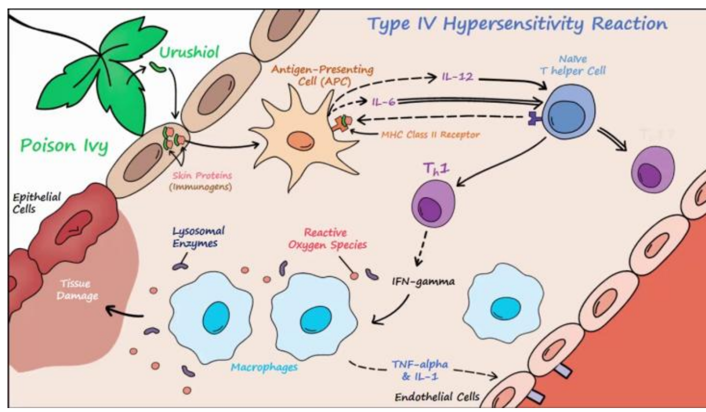
↓

T helper 1 and memory T cells develop.

↓

2nd exposure, Th1 cells secrete IFNγ/recruit macrophages.

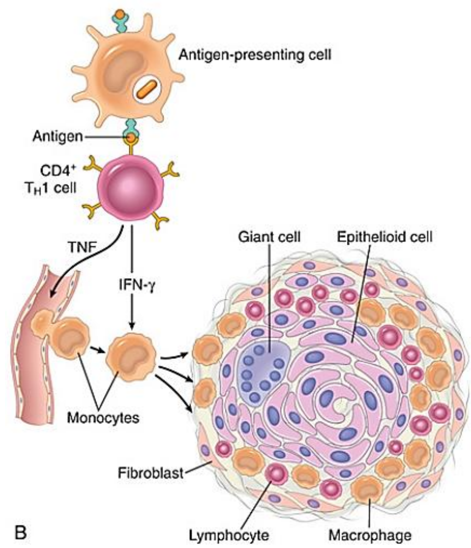
Contact dermatitis



Contact dermatitis will be characterized by pruritis, redness, swelling and sometimes blistering, or scaling or ulceration.

Granulomas is a type IV hypersensitivity

- Granulomatous inflammation (chronic inflammation, Th-mediated type IV hypersensitivity). Characterized by
  - Accumulation of activated macrophage, termed epithelioid cells.
  - IFN-γ can induced the epithelioid cells to form multinucleate giant cells.
  - A collar of lymphocytes typically surround the epithelioid cells (together forming a granuloma)



Detecting Type IV hypersensitivity: Example: tuberculin skin test

Mycobacterium tuberculosis

Positive reaction

- *Mycobacterium tuberculosis* causes tuberculosis disease.
- Tuberculin test is used to detect exposure to M. tuberculosis via injecting a purified protein derivative (PPD) intradermally. PPD is derived from mycobacterium cell wall.
- After 2-3 days, a positive skin-test reaction indicates that the individual has sensitized Th1 cells specific for the TB purified protein (BCG vaccine, active or latent infection).
- Now, interferon gamma release serum testing (e.g: QuantiFERON-TB gold) has higher accuracy and doesn't need multiple visit.

Penicillin can induce all 4 hypersensitivity types.

TABLE 15-4 Penicillin-induced hypersensitivity reactions

Type of reaction	Antibody or lymphocyte induced	Clinical manifestations
I	IgE	Urticaria, systemic anaphylaxis
II	IgM, IgG	Hemolytic anemia
III	IgG	Serum sickness, glomerulonephritis
IV	T cells	Contact dermatitis

Management Strategies

Type I hypersensitivity

1- Avoid exposure to the triggering allergens.

2- Desensitization or hyposensitization.

3- Antihistamines, inhalation corticosteroids.

4- Therapeutic anti-IgE antibodies known by omalizumab (Xolair).

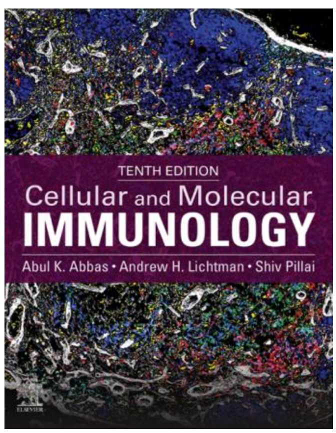
5- Epinephrine, or Beta 2 agonists (like albuterol for asthma).

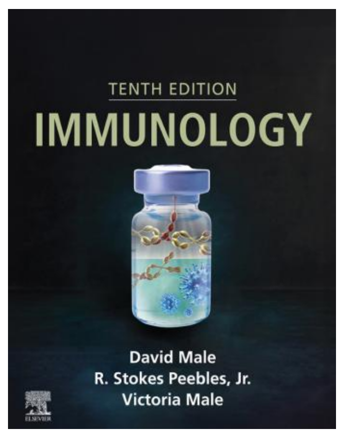
Management strategies (cont.)

Type II-IV hypersensitivity

- Treatment course varies according to the clinical presentation.
- In addition to alleviating the impact of tissue damage, other treatment strategies can include systemic glucocorticoids, NSAIDs, and avoiding/removing the insulting trigger.

Resources


TENTH EDITION
Cellular and Molecular
IMMUNOLOGY
Abul K. Abbas • Andrew H. Lichtman • Shiv Pillai


TENTH EDITION
IMMUNOLOGY
David Male
R. Stokes Peebles, Jr.
Victoria Male